Josiah Lo

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WORK EXPERIENCE

EMCORE Corporation, Quartz MEMS Division — *Mechanical Engineer Intern* **Jan 2026 - Present**

- Enabled $716K in inertial measurement unit sales by developing a Python script to rapidly analyze test history across ~2,000 scrapped sensors, identifying 276 viable accelerometers and 375 gyroscopes ($330K in parts) validated against a customer's new 30+ test specifications
- Designed and executed a 40-simulation FEA study to evaluate 8 die package designs of the MEMS accelerometer via coupled thermal-structural-modal-vibration simulations to compare performance under package stress, thermal load, and vibration—advancing the best simulated design to a pilot production run
- Traced MEMS accelerometer yield failures to moisture contamination in the hermetic sealing process by isolating a discrete -15°C frequency shift in 50 of 179 units with a MATLAB data pipeline, reproducing the observed frequency shift of the quartz resonator with a structural-modal Finite Element Analysis in Ansys, and verifying the contamination via mass spectroscopy
- Saved $7K in parts cost and preserved $147K in shipped IMU revenue by selecting units for a reworking process (drill, vacuum-bake, reseal) that recovered 77% of moisture-contaminated accelerometers
- Validated IMU for Lockheed Martin Space delta qualification shock and vibration profiles via Ansys shock-response-spectrum and vibration simulations, and compiled technical report for customer verification

EDUCATION

University of California, Berkeley **Expected Graduation May 2027**

Bachelor of Science in Mechanical Engineering **GPA: 4.000**

Dean's List - College of Engineering, All Semesters

Relevant Coursework: Behavior of Materials, Mechanics, Circuit Design, 3D Modeling, Manufacturing and GD&T, Finite Element Analysis, Feedback Control Systems

RELEVANT PROJECTS

UC Berkeley Solar Vehicle Team — *Project Lead* **Oct 2025 - Present**

- Evaluated composite laminate suspension box for required safety factor using Ansys ACP FEA — Improved design to increase safety factor from 0.09 to 1.03 while limiting weight increase to 8.9%
- Drafted drawing for composite parts and designed manufacturing plan for waterjetting and epoxy bonding
- Led design and manufacturing efforts for suspension box team, coordinating assembly and integration with adjacent vehicle subsystem teams
- Redesigned roll cage in SolidWorks for strength-to-weight ratio, verified safety factor and max deflection via Ansys structural FEA, and selected roll cage material to optimize for strength, cost, and manufacturability

6-Axis Surgical Robot — *Personal Project* **Dec 2025 - May 2026**

- Designed robotic arm in SolidWorks and fabricated components with 3D printing, including a custom 2-stage 15:1 cycloidal gearbox at the base joint
- Routed and integrated wiring through the arm to maintain 6-axis motion clearance with 20-inch reach radius

Smart Window — *UC Berkeley Electronics Project* **Nov 2025 - Dec 2025**

- Designed an autonomous smart window integrating motorized actuation, locking, and polymer-dispersed liquid crystal tint control, with temperature, smoke, and noise sensing to provide comfort, safety, and privacy
- Designed custom control circuitry to interface three ESP32 microcontrollers with sensors, motors, and actuators, accounting for power distribution and signal requirements — programming them in Python to enable fully autonomous operation, including Wi-Fi communication, distributed sensor data sharing, and closed-loop control logic
- Modeled and 3D-printed enclosures for electronics, optimizing wiring clearance and component accessibility

SKILLS

Software and Programming: Ansys Mechanical, SolidWorks, Creo, AutoCAD, LTspice, MATLAB

Fabrication & Testing: 3D Printing, Mill, Lathe, GD&T per ASME Y14.5, Soldering, Lock-in Amplifier, PLL